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1.1 Lab 1

1.1.1 Author

package lab1;

```
public class Author {  
    private String name;  
    private String email;  
    private String gender;  
  
    public Author() {  
  
    }  
  
    public Author(String name, String email, String gender) {  
        this.name = name;  
        this.email = email;  
        this.gender = gender;  
    }  
  
    public String getName() {  
        return name;  
    }  
  
    public void setName(String name) {  
        this.name = name;  
    }  
  
    public String getEmail() {  
        return email;  
    }  
  
    public void setEmail(String email) {  
        this.email = email;  
    }  
  
    public String getGender() {  
        return gender;  
    }  
  
    public void setGender(String gender) {  
        this.gender = gender;  
    }  
  
    @Override  
    public String toString(){  
        return name + " (" + gender + "), " + email;  
    }  
}
```

```

    }
}
1.1.2 TestAuthor
package lab1;

import java.util.Scanner;

public class TestAuthor {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        /* Nhập từ bàn phím */
        System.out.print("Nhập tên tác giả: ");
        String name = scanner.nextLine();

        System.out.print("Nhập email: ");
        String email = scanner.nextLine();

        System.out.print("Nhập giới tính (male/female): ");
        String gender = scanner.nextLine();
        /* Tao đối tượng Author */
        Author author = new Author(name, email, gender);

        /* Hiện thị thông tin */
        System.out.println("Thông tin tác giả:");
        System.out.println(author);

        scanner.close();
    }
}

```

1.2 Lab 2

1.2.1 Circle

```

package lab2;

public class Circle {
    private double radius = 1.0;
    private String color = "red";

    public Circle() {}

    public Circle(double radius) {
        this.radius = radius;
    }

    public double getRadius() {
        return radius;
    }
}

```

```

public void setRadius(double radius) {
    this.radius = radius;
}

public String getColor() {
    return color;
}

public void setColor(String color) {
    this.color = color;
}

public double getArea() {
    return Math.PI * radius * radius;
}

public String toString() {
    return "Radius (" + radius + "), color (" + color + ")";
}
}

```

1.2.2 Cylinder

```

package lab2;

```

```

public class Cylinder extends Circle {
    private double height = 1.0;

    public Cylinder() {
        super();
    }

    public Cylinder(double radius) {
        super(radius);
    }

    public Cylinder(double radius, double height) {
        super(radius);
        this.height = height;
    }

    public double getHeight() {
        return height;
    }

    public void setHeight(double height) {
        this.height = height;
    }

    public double getVolume() {

```

```

        return getArea() * height;
    }

    public double getLateralSurfaceArea() {
        return 2 * Math.PI * getRadius() * height;
    }
}

1.2.3 Main
package lab2;

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Nhập bán kính hình tròn: ");
        double radius = scanner.nextDouble();

        System.out.print("Nhập chiều cao hình trụ: ");
        double height = scanner.nextDouble();

        scanner.nextLine();
        System.out.print("Nhập màu sắc: ");
        String color = scanner.nextLine();

        Cylinder cylinder = new Cylinder(radius, height);
        cylinder.setColor(color);

        System.out.println("\nThông tin hình tròn:");
        System.out.println(cylinder.toString());
        System.out.println("Diện tích hình tròn: " + cylinder.getArea());

        System.out.println("\nThông tin hình trụ:");
        System.out.println("Chiều cao: " + cylinder.getHeight());
        System.out.println("Thể tích hình trụ: " + cylinder.getVolume());
        System.out.println("Diện tích mặt xung quanh hình trụ: " +
cylinder.getLateralSurfaceArea());
    }
}

```

1.3 Lab 3

1.3.1 HocSinh

```

abstract public class HocSinh {
    public String Ma;
    public String Ten;
    abstract public void Nhap();
    abstract public void Xuat();
    abstract public float DiemTB();
}

```

```
        abstract public String XepLoai();  
    }
```

1.3.2 THCS

```
import java.util.Scanner;
```

```
public class THCS extends HocSinh {  
    public float DiemToan = 0.0f;  
    public float DiemNguVan = 0.0f;  
    public float DiemAnhVan = 0.0f;
```

```
    public THCS() {
```

```
    }
```

```
    @Override
```

```
    public void Nhap() {
```

```
        Scanner sc = new Scanner(System.in);  
        System.out.print("Nhap ma hoc sinh: ");  
        Ma = sc.nextLine();
```

```
        System.out.print("Nhap ten hoc sinh: ");  
        Ten = sc.nextLine();
```

```
        System.out.print("Nhap diem toan: ");  
        DiemToan = sc.nextFloat();
```

```
        System.out.print("Nhap diem ngu van: ");  
        DiemNguVan = sc.nextFloat();
```

```
        System.out.print("Nhap diem anh van: ");  
        DiemAnhVan = sc.nextFloat();
```

```
    }
```

```
    @Override
```

```
    public void Xuat() {
```

```
        System.out.println("Ma: " + Ma + ", Ten: " + Ten + ", Diem toan: " +  
DiemToan +  
                                ", Diem ngu van: " + DiemNguVan + ", Diem anh van: " +  
DiemAnhVan +  
                                ", Diem trung binh: " + DiemTB() + ", Xep loai: " +  
XepLoai());  
    }
```

```
    @Override
```

```
    public float DiemTB() {
```

```
        return (DiemToan + DiemNguVan + DiemNguVan )/3;
```

```
    }
```

```

@Override
public String XepLoai() {
    float dtb = DiemTB();
    if(dtb < 5.0f) return "Yeu";
    else if(dtb >= 5.0f && dtb < 6.5f) return "Trung binh";
    else if(dtb >= 6.5f && dtb < 8.0f) return "Kha";
    else return "Gioi";
}
}

```

1.3.3 TieuHoc

```

import java.util.Scanner;
public class TieuHoc extends HocSinh{
    public float DiemToan = 0.0f;
    public float DiemTiengViet = 0.0f;

    public TieuHoc() {
    }
}

```

```

@Override
public void Nhap() {
    Scanner sc = new Scanner(System.in);
    System.out.print("Nhap ma hoc sinh: ");
    Ma = sc.nextLine();

    System.out.print("Nhap ten hoc sinh: ");
    Ten = sc.nextLine();

    System.out.print("Nhap diem toan: ");
    DiemToan = sc.nextFloat();

    System.out.print("Nhap diem tieng viet: ");
    DiemTiengViet = sc.nextFloat();
}

```

```

@Override
public void Xuat() {
    System.out.println("Ma: " + Ma + ", Ten: " + Ten + ", Diem toan: " +
    DiemToan +
    ", Diem tieng viet: " + DiemTiengViet +
    ", Diem trung binh: " + DiemTB() +
    ", Xep loai: " + XepLoai());
}

```

```

@Override

```

```

public float DiemTB() {
    return (DiemToan + DiemTiengViet)/2;
}

```

```

@Override
public String XepLoai() {
    float dtb = DiemTB();
    if(dtb < 5.0f) return "Yeu";
    else if(dtb >= 5.0f && dtb < 6.5f) return "Trung binh";
    else if(dtb >= 6.5f && dtb < 8.0f) return "Kha";
    else return "Gioi";
}

```

```

}

```

1.3.4 Main

```

import java.util.Scanner;
import java.util.ArrayList;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ArrayList<HocSinh> danhSach = new ArrayList<>();
        System.out.print("Nhap so luong hoc sinh: ");
        int soLuong = sc.nextInt();
        sc.nextLine();
        for (int i = 0; i < soLuong; i++) {
            System.out.println("\nNhap thong tin hoc sinh thu " + (i + 1) + ":");
            System.out.print("Chon loai hoc sinh(1. Tieu hoc, 2. THCS): ");
            int loai = sc.nextInt();
            sc.nextLine();
            HocSinh hs;
            if (loai == 1) {
                hs = new TieuHoc();
            } else if (loai == 2) {
                hs = new THCS();
            } else {
                System.out.println("Lua chon khong hop le. Bo qua hoc sinh nay.");
                continue;
            }
            hs.Nhap();
            danhSach.add(hs);
        }
        System.out.println("\n----- DANH SACH HOC SINH -----");
        for (HocSinh hs : danhSach) {
            hs.Xuat();
        }
    }
}

```


1.4 OOP_2

1.4.1 Product

```
import java.util.Date;
import java.util.Scanner;
import java.text.SimpleDateFormat;
public class Product {
    private String ID;
    private String Name;
    private float Price;
    private Date MgfDate;
    private Date ExpiryDate ;

    public String getID() {
        return ID;
    }

    public void setID(String iD) {
        ID = iD;
    }

    public String getName() {
        return Name;
    }

    public void setName(String name) {
        this.Name = name;
    }

    public float getPrice() {
        return Price;
    }

    public void setPrice(float price) {
        this.Price = price;
    }

    public Date getMgfDate() {
        return MgfDate;
    }

    public void setMgfDate(Date mgfDate) {
        this.MgfDate = mgfDate;
    }

    public Date getExpiryDate() {
        return ExpiryDate;
    }
}
```

```

public void setExpiryDate(Date expiryDate) {
    this.ExpiryDate = expiryDate;
}

public Product() {
}

public Product(String iD, String name) {
    this.ID = iD;
    this.Name = name;
}

public void InputProduct() {
    Scanner scanner = new Scanner(System.in);
    System.out.print("Nhap ID san pham: ");
    this.ID = scanner.nextLine();
    System.out.print("Nhap ten san pham: ");
    this.Name = scanner.nextLine();
    System.out.print("Nhap gia san pham: ");
    this.Price = scanner.nextFloat();
    scanner.nextLine();
    SimpleDateFormat formatter = new SimpleDateFormat("dd/MM/yyyy");
    try {
        System.out.print("Nhap ngay san xuat (dd/MM/yyyy): ");
        String mgfDateStr = scanner.nextLine();
        this.MgfDate = formatter.parse(mgfDateStr);
        System.out.print("Nhap han su dung (dd/MM/yyyy): ");
        String expiryDateStr = scanner.nextLine();
        this.ExpiryDate = formatter.parse(expiryDateStr);
    } catch (Exception e) {
        System.out.println("Loi: Dinh dang ngay khong hop le!");
    }
}

public void ShowProduct() {
    System.out.println("Ma san pham: " + this.ID);
    System.out.println("Ten san pham: " + this.Name);
    System.out.println("Gia: " + this.Price);
    System.out.println("Ngay san xuat: " + this.MgfDate);
    System.out.println("Han su dung: " + this.ExpiryDate);
}
}

```

1.4.2 Supplier

```
import java.util.Scanner;
```

```
public class Supplier {
```

```
private String ID;  
private String Name;  
private String PhNo;  
private String Address;  
  
public String getID() {  
    return ID;  
}  
  
public void setID(String iD) {  
    ID = iD;  
}  
  
public String getName() {  
    return Name;  
}  
  
public void setName(String name) {  
    this.Name = name;  
}  
  
public String getPhNo() {  
    return PhNo;  
}  
  
public void setPhNo(String phno) {  
    this.PhNo = phno;  
}  
  
public String getAddress() {  
    return Address;  
}  
  
public void setAddress(String address) {  
    this.Address = address;  
}  
  
public Supplier() {  
  
}  
public Supplier(String iD, String name, String phno) {  
    this.ID = iD;  
    this.Name = name;  
    this.PhNo = phno;  
}  
  
public void InputSupplier() {  
    Scanner scanner = new Scanner(System.in);
```

```

        System.out.print("Nhap ID nha cung cap: ");
        this.ID = scanner.nextLine();

        System.out.print("Nhap ten nha cung cap: ");
        this.Name = scanner.nextLine();

        System.out.print("Nhap so dien thoai nha cung cap: ");
        this.PhNo = scanner.nextLine();

        System.out.print("Nhap dia chi nha cung cap: ");
        this.Address = scanner.nextLine();

    }

    public void ShowSupplier() {
        System.out.println("Ma nha cung cap: " + this.ID);
        System.out.println("Ten nha cung cap: " + this.Name);
        System.out.println("So dien thoai" + this.PhNo);
        System.out.println("Đia chi: " + this.Address);
    }
}

```

1.4.3 Main

```

import java.util.ArrayList;
import java.util.Scanner;
public class Main {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);
        System.out.print("Nhap so luong danh sach: ");
        int n = scanner.nextInt();
        Product[] dssp = new Product[n];

        for(int i = 0; i < n; i++) {
            System.out.println("Nhap thong tin san pham thu " + (i + 1) + ": ");
            dssp[i] = new Product();
            dssp[i].InputProduct();
            System.out.println();
        }
        for(int i = 0; i < n; i++) {
            dssp[i].ShowProduct();
            System.out.println();
        }

        Supplier[] dsnsx = new Supplier[n];

        for(int i = 0; i < n; i++) {
            System.out.println("Nhap thong tin nha san xuat thu " + (i + 1) + ": ");

```

```
dsnsx[i] = new Supplier();
dsnsx[i].InputSupplier();
System.out.println();
}

for(int i = 0; i < n; i++) {
    dsnsx[i].ShowSupplier();
    System.out.println();
}
}
```